

HANOI UNIVERSITY OF SCIENCE AND TECHNOLOGY

School of Information and communications technology

Software Requirement Specification

Version 1.2

An Internet Media Store

Subject: E-commerce software
development

Group 1

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1 Introduction

1.1 Objective

The purpose of this Software Requirements Specification (SRS) document is to clearly outline the functionalities and needs of the AIMS software project. It acts as a blueprint for everyone involved in the development process.

The intended audience for this SRS includes various teams:

- **Developers and Designers:** The SRS provides them with a roadmap to design and build the software according to the specified requirements.
- **Testers:** The document allows them to create test cases to ensure the final product meets the outlined functionalities.
- **Project Managers:** The SRS helps them estimate timelines, costs, and resources needed for the project.

1.2 Scope

AIMS Project (An Internet Media Store)

The AIMS Project delivers a robust desktop e-commerce software solution for efficient product management, order processing, and customer interaction. Designed for administrators, product managers, and customers, AIMS empowers businesses to manage their online stores effectively.

- **Streamlined Product Management:** Product managers can effortlessly add, edit, and manage product information, ensuring accurate data and efficient inventory control.
- **Seamless Order Processing:** Customers can browse a user-friendly interface, search for products, and build shopping carts. The system integrates with VNPay for secure online payment processing, streamlining the checkout experience.
- **Enhanced Customer Experience:** 24/7 operation and high performance guarantee continuous service availability. Customers receive order confirmations and can track order history, fostering trust and convenience.
- **Scalability and Security:** AIMS is designed to handle up to 1,000 concurrent users with minimal performance impact. Robust security measures protect user accounts and transaction data.

1.3 Glossary

<i>No</i>	<i>Term</i>	<i>Explanation</i>	<i>Example</i>	<i>Note</i>
1	Media product	A category of product specifically referring to physical media	Books, CDs, DVDs	
2	Role	A user designation defining access and permissions within the software	Administrator, product manager	
3	Product manager	A user role responsible for managing product information within the software (adding, editing, deleting)		
4	Administrator	A user role with permissions to manage users (create, update, delete)		
5	VAT	A tax added to the product price by the government		The specific VAT can vary depending on the country/region
6	Rush order delivery	A special delivery option where customers receive their orders significantly faster than usual		Limited by area and product type
7	Out of stock	The state of a product that is no longer available for purchase		
8	Cart	A temporary virtual holding area for products a customer chooses before checkout		

<i>No</i>	<i>Term</i>	<i>Explanation</i>	<i>Example</i>	<i>Note</i>
9	Order	A customer's confirmed selection of products they intend to purchase		
10	Invoice	A detailed receipt showing the purchased items, quantities, prices, total amount, and delivery information		
11	Transaction	The successful completion of a financial exchange between the customer and the software		Usually via credit card

1.4 References

- Lecture slides
- AIMS-ProblemStatement-v2.0

2 Overall Description

2.1 Survey

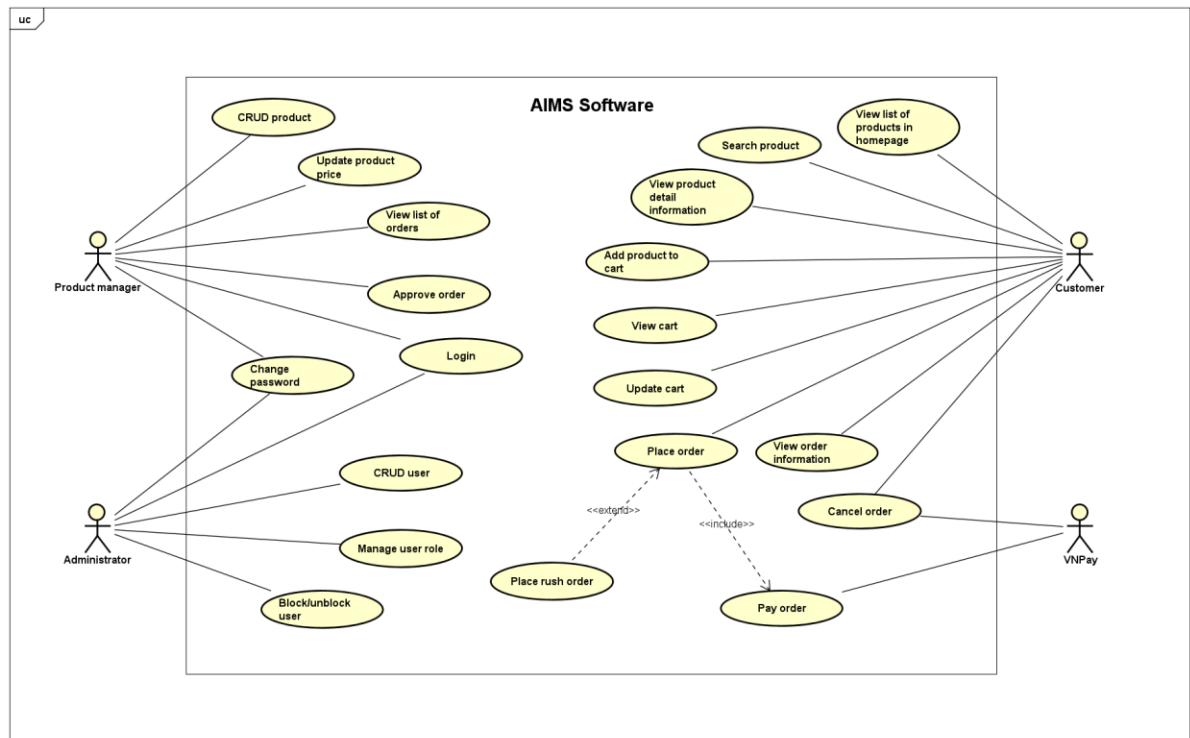
Software Overall Description

The software is an e-commerce platform designed for selling physical media products (e.g., books, CDs, DVDs) called AIMS (An Internet Media Store).

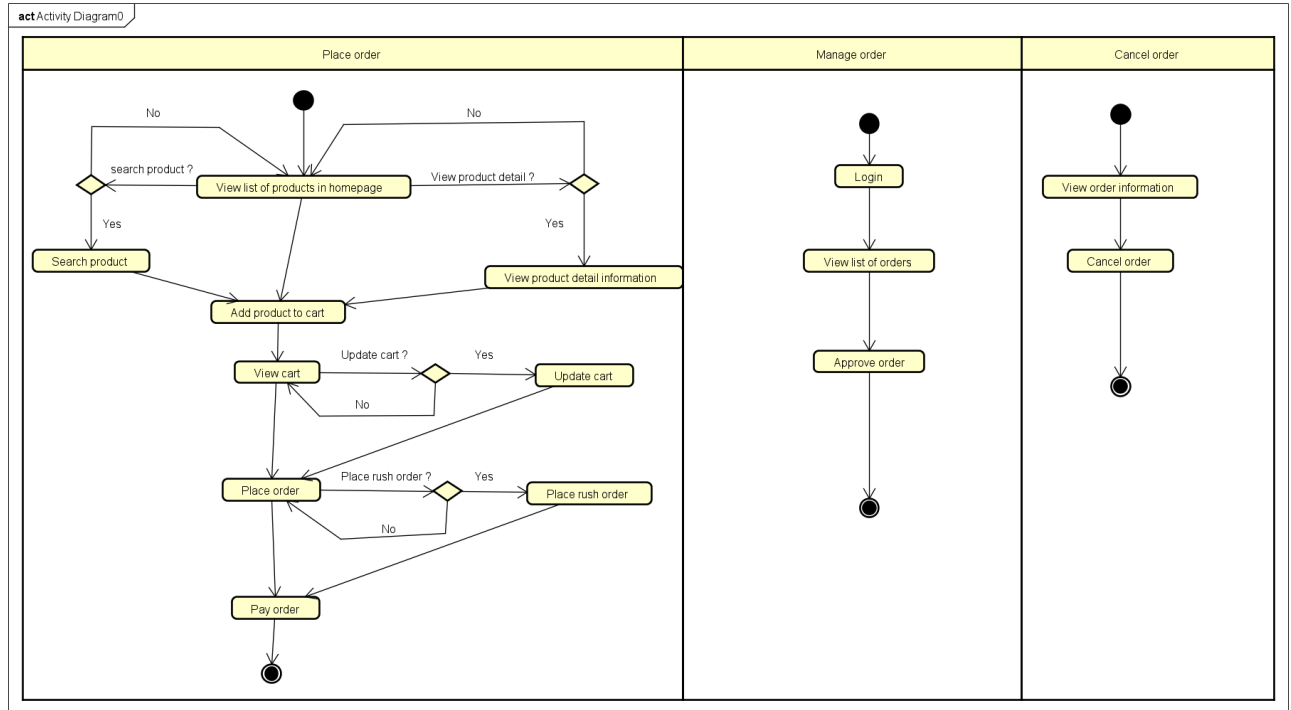
List of Actors

- **Customer:** The primary user who interacts with the software to browse products, search for specific items, add products to a cart, choose a delivery method (including potentially rush order delivery), complete the checkout process with secure payment via VNPay
- **VNPay:** An external third-party payment processing service that securely handles online transactions for credit card payments made by customers within the AIMS software
- **Product manager:** A user with specific permissions within the software who can manage products and approve or reject order
- **Administrator:** A user with the highest level of permissions within the software who can manage user accounts and user roles

2.2 Overall requirements



2.3 Business process



3 Detailed Requirements

3.1 Use case “Place order”

Use Case “Place Order”

1. Use case code

UC001

2. Brief Description

This use case describes the interaction between customers and AIMS software when the customer wishes to place order.

3. Actors

3.1 Customer

4. Preconditions

There is at least one item in the cart.

5. Basic Flow of Events

1. Customer requests to place order in the cart
2. AIMS software checks the availability of products in the cart
3. AIMS software displays the form of delivery information with order information
4. Customer enters and submits delivery information (see Table 2)
5. AIMS software checks the validity of the delivery information
6. AIMS software calculates and displays order and shipping fees (see Table 3)
7. The customer asks to pay order
8. The AIMS software calls UC “Pay order”
9. The AIMS software creates a new order
10. The AIMS software makes the cart empty
11. The AIMS software sends email about the order notification and information
12. The AIMS software displays the successful order notification, the order and the transaction information (see Table 4)

6. Alternative flows

Table 1-Alternative flows of events for UC Place order

No	Location	Condition	Action	Resume location
1.	At Step 3	If the products are not available	▪ The AIMS software notifies that the products in the cart are not	Use case ends

			available and stay at the use case “View cart”	
2.	At Step 6	If the delivery info is invalid	AIMS software notifies that the delivery info is invalid (blank or wrong format)	Resumes at Step 3
3.	At Step 6	If the user chooses to place a rush order	AIMS software inserts use case “Place rush order”	Resumes at Step 7
4.	At Step 9	If the order payment is not successful or goes back from payment		Resumes at Step 6

7. Input data

Table 2-Input data of delivery information

No	Data fields	Description	Mandatory	Valid condition	Example
1.	Recipient name		Yes		Do Minh Hieu
2.	Email		Yes	- Contains ‘@’ - No space	mhieu@gmail.com
3.	Phone number		Yes	10 digits	0987654321
4.	Province	Choose from a list	Yes		Hanoi
5.	Delivery address		Yes		12, 34 Alley of Tran Thai Tong street, Cau Giay district

8. Output data

Table 3-Output data of order information and shipping fee

No	Data fields	Description	Display format	Example
1.	Title	Title of a media product		DVD Phim Vượt ngục

2.	Price	Price of the corresponding product	- Comma for thousands separator - Positive integer - Right alignment	123,000
3.	Quantity	Quantity of the corresponding media	- Positive integer - Right alignment	2
4.	Amount	Total money of the corresponding media	- Comma for thousands separator - Positive integer - Right alignment	246,000
5.	Subtotal	Total amount of all products in the order	- Comma for thousands separator - Positive integer - Right alignment	2,316,600
6.	Shipping fee			30,000

(The rows with green shading are repeated for all media products in the cart)

Table 4-Output data of general information of order and transaction info

No	Data fields	Description	Display format	Example
1.	Customer name			Do Minh Hieu
2.	Phone number			0987654321
3.	Province			Hanoi
4.	Address			12, 34 Alley of Tran Thai Tong street, Cau Giay district
5.	Total amount		- Right alignment - Vietnamese currency (VNĐ) - Vietnamese locale	1.200.000 VNĐ
6.	Transaction ID			
7.	Transaction content			
8.	Transaction date		dd/mm/yyyy	05/10/2023

9. Postconditions

A new order is created, and its information is sent via email to the customer or nothing happens if payment is not successful

3.2 Use case “Pay order”

Use Case “Pay Order”

1. Use case code

UC002

2. Brief Description

This use case describes the interactions between the AIMS software with the customer and VNPay when the customer desires to pay order.

3. Actors

3.1 Customer

3.2 VNPay

4. Preconditions

The AIMS software has caculated the delivery fee.

5. Basic Flow of Events

1. AIMS software displays the invoice (see Table 2)
2. Customer asks to pay the invoice
3. AIMS software redirects to VNPay with payment information
4. VNPay notifies the transaction result
5. AIMS software saves the payment transaction

6. Alternative flows

Table 1-Alternative flows of events for UC Pay order

No	Location	Condition	Action	Resume location
1.	At Step 5	If the customer cancels the payment transaction	▪	Resumes at Step 1

7. Output data

Table 2-Output data of invoice

No	Data fields	Description	Display format	Example
1.	Title	Title of a media product		DVD Phim Vượt ngục
2.	Price	Price of the corresponding media product	- Comma for thousands separator - Positive integer - Right alignment	123,000
3.	Quantity	Quantity of the corresponding media	- Positive integer - Right alignment	2
4.	Amount	Total money of the corresponding media	- Comma for thousands separator - Positive integer - Right alignment	246,000
5.	Subtotal Before VAT	Total price of products in the cart before VAT	- Comma for thousands separator - Positive integer - Right alignment	2,106,000
6.	Subtotal	Total price of products in the cart with VAT		2,316,600
7.	Shipping fees			30,000
8.	Total	Sum of subtotal and shipping fees		2,346,600
9.	Currency			VND
10.	Name			Do Minh Hieu
11.	Phone number			0987654321
12.	Province			Hanoi
13.	Address			12, 34 Alley of Tran Thai Tong street, Cau Giay district
14.	Shipping instructions			

(The rows with green shading are repeated for all media products in the invoice)

8. Postconditions

The customer has successfully paid for the order, or no changes have been made to the customer's credit account if the payment process was unsuccessful.

3.3 Use case *"Place rush order"*

Use Case **"Place Rush Order"**

1. Use case code

UC003

2. Brief Description

This use case describes the interaction between customers and AIMS software when the customer wishes to place rush order.

3. Actors

3.1 Customer

4. Preconditions

Customer has selected rush order delivery

5. Basic Flow of Events

1. The AIMS software checks whether the delivery address supports this service and if any products are eligible
2. Software requests additional rush order delivery information from the customer and provides other information
3. Customer enters and submit additional rush order delivery information (see Table 2)
4. Software recalculates the delivery fees and updates the corresponding invoice (see Table 3)

6. Alternative flows

Table 1-Alternative flows of events for UC Place order

No	Location	Condition	Action	Resume location
1.	At Step 2	If no products are eligible or the delivery address doesn't support rush order delivery	▪ The software prompts the customer to update the delivery information or delivery method	Use case ends

7. Input data

Table 2-Input data of additional rush order delivery information

No	Data fields	Description	Mandatory	Valid condition	Example
1.	Delivery time	Choose from a list	Yes		10:00 AM 14/03/2024
2.	Delivery instructions		No		

8. Output data

Table 3-Output data of order information and shipping fee

No	Data fields	Description	Display format	Example
1.	Title	Title of a media product		DVD Phim Vượt ngục
2.	Price	Price of the corresponding media product	- Comma for thousands separator - Positive integer - Right alignment	123,000
3.	Quantity	Quantity of the corresponding media	- Positive integer - Right alignment	2
4.	Amount	Total money of the corresponding media	- Comma for thousands separator - Positive integer - Right alignment	246,000
5.	Subtotal	Total amount of all products in the order	- Comma for thousands separator - Positive integer - Right alignment	2,316,600
6.	Shipping fee			30,000

(The rows with green shading are repeated for all media products in the cart)

9. Postconditions

The customer has successfully selected the rush order delivery option, the delivery fee has been updated to reflect the rush order status, or no changes have been made to the customer's invoice status.

4 Supplementary specification

4.1 Functionality

4.1.1 User authentication

The system shall require user authentication (login) with username and password for administrators and product managers.

4.1.2 Data management

The system shall validate all user inputs to ensure data integrity and prevent invalid data entry. This includes:

- Validating product information based on media type
- Ensuring product prices are within the allowed range (30-150% of product value).
- Checking the validity of the delivery information
- Checking delivery address supportability and product eligibility for rush order service.
- Verifying sufficient inventory for ordered items.

4.2 Usability

- The system should be designed with an intuitive user interface to minimize the learning curve for new users.
- The system should provide clear and informative messages to guide users during interactions.

4.3 Reliability

- The system should operate continuously for a minimum of 300 hours without failure
- The system should resume normal operation within a maximum of 1 hour after an incident

4.4 Performance

- The system should have a maximum response time of 2 seconds under normal conditions.
- The system should have a maximum response time of 5 seconds during peak hours.

4.5 Supportability

- The system should be developed using well-documented and modular code for ease of maintenance and future enhancements.
- The system should provide comprehensive logs for troubleshooting and error identification.

4.6 Other requirements

4.6.1 Database System

The system database shall be robust and scalable to handle product information, user data, orders, and inventory management.

4.6.2 Used Technology

The system shall integrate with VNPay for secure online payment processing via credit card.